Chapter 3, Section 6

James Lee

April 19, 2026

1. Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{F}', \mathcal{F}'' \in \mathfrak{Mod}(X)$. An *extension* of $\mathcal{F}''$ by $\mathcal{F}'$ is a short exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

in $\mathfrak{Mod}(X)$. Two extensions are *isomorphic* if there is an isomorphism of the short exact sequences, inducing the identity maps on $\mathcal{F}'$ and $\mathcal{F}''$. Given an extension as above consider the long exact sequence arising from $\operatorname{Hom}(\mathcal{F}'', \cdot)$, in particular the map

$$\delta : \operatorname{Hom}(\mathcal{F}'', \mathcal{F}'') \to \operatorname{Ext}^1(\mathcal{F}'', \mathcal{F}'),$$

and let $\xi \in \operatorname{Ext}^1(\mathcal{F}'', \mathcal{F}')$ by $\delta(1_{\mathcal{F}''})$. Show that this process gives a one-to-one correspondence between isomorphism classes of extensions of $\mathcal{F}''$ by $\mathcal{F}'$, and elements of the group $\operatorname{Ext}^1(\mathcal{F}'', \mathcal{F}')$.

Proof. Let $0 \to \mathcal{F}' \to \mathcal{I}^\bullet$ be an injective resolution of $\mathcal{F}'$. An element of the group $\xi \in \operatorname{Ext}^1(\mathcal{F}'', \mathcal{F}')$ gives a morphism $\xi : \mathcal{F}'' \to \mathcal{I}^1$ such that

$$\begin{tikzcd}
\mathcal{F}'' \arrow[d, "\xi"'] \arrow[dr, "0"] & \\
\mathcal{I}^1 \arrow[r, "\psi"] & \mathcal{I}^2
\end{tikzcd}$$

commutes, i.e., $\xi \in \operatorname{Hom}(\mathcal{F}'', \mathcal{I}^0/\mathcal{F}')$. From now, denote this subsheaf of $\mathcal{I}^1$ as $\mathcal{J} = \mathcal{I}^0/\mathcal{F}'$. The pull-back diagram of ξ and ψ gives the following commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}' \arrow[r] \arrow[d, equal] & \mathcal{I}^0 \times_{\mathcal{J}^1} \mathcal{F}'' \arrow[r, dashed] \arrow[d, dashed] & \mathcal{F}'' \arrow[r] \arrow[d, "\xi"'] & 0 \\
0 \arrow[r] & \mathcal{F}' \arrow[r] & \mathcal{I}^0 \arrow[r, "\psi"] & \mathcal{J} \arrow[r] & 0
\end{tikzcd},$$

so the top row is an extension of $\mathcal{F}''$ by $\mathcal{F}'$. If we are given another representative $\bar{\xi}$ of ξ, then $\bar{\xi} - \xi = j \circ \psi$ for some $\psi \in \operatorname{Hom}(\mathcal{F}'', \mathcal{I}^0)$, and repeating the construction above for $j \circ \psi$ gives the trivial extension $\mathcal{F}' \times \mathcal{F}''$ of $\mathcal{F}''$ by $\mathcal{F}'$, so this construction is well-defined. Because $\operatorname{Hom}(\mathcal{F}'', \cdot)$ is a covariant δ-functor, there is a commutative diagram

$$\begin{tikzcd}
\operatorname{Hom}(\mathcal{F}'', \mathcal{F}'') \arrow[r, "\delta"] \arrow[d] & \operatorname{Ext}^1(\mathcal{F}'', \mathcal{F}') \arrow[d, equal] \\
\operatorname{Hom}(\mathcal{F}'', \mathcal{J}) \arrow[r] & \operatorname{Ext}^1(\mathcal{F}'', \mathcal{F}')
\end{tikzcd}$$

which ensures $\delta(1_{\mathcal{F}''}) = \xi$. Hence, this process is surjective.

To prove injectivity, we show $\mathcal{F} \cong \mathcal{I}^0 \times_{\mathcal{J}^1} \mathcal{F}''$. Let $\alpha : \mathcal{F} \to \mathcal{F}''$ be the epimorphism from the extension. By injectivity of $\mathcal{I}^0$, there exists a morphism $\beta : \mathcal{F} \to \mathcal{I}^0$ that extends $\mathcal{F}' \to \mathcal{I}^0$, which induces a morphism $\gamma : \mathcal{F}'' \to \mathcal{I}^1$ making the diagram commute

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}' \arrow[r] \arrow[d, equal] & \mathcal{F} \arrow[r] \arrow[d, "\beta"'] & \mathcal{F}'' \arrow[r] \arrow[d, "\gamma"'] & 0 \\
0 \arrow[r] & \mathcal{F}' \arrow[r] & \mathcal{I}^0 \arrow[r] & \mathcal{J} \arrow[r] & 0
\end{tikzcd}$$

The pull-back $\mathcal{I}^0 \times_{\mathcal{J}^1} \mathcal{F}''$ is unique up to unique isomorphism, so it remains to show γ is a representative of ξ. Again, $\operatorname{Hom}(\mathcal{F}'', \cdot)$ is a covariant δ-functor with $\operatorname{Hom}(\mathcal{F}'', \mathcal{F}'') \to \operatorname{Hom}(\mathcal{F}'', \mathcal{J})$ given by composition with γ, so γ is a representative of ξ. $\square$

3. Let X be a Noetherian scheme, and let $\mathcal{F}, \mathcal{G} \in \mathfrak{Mod}(X)$.

(a) If $\mathcal{F}, \mathcal{G}$ are both coherent, then $\mathbf{Ext}^i(\mathcal{F}, \mathcal{G})$ is coherent, for all $i \geq 0$.

(b) If $\mathcal{F}$ is coherent and $\mathcal{G}$ is quasi-coherent, then $\mathbf{Ext}^i(\mathcal{F}, \mathcal{G})$ is quasi-coherent, for all $i \geq 0$.

Proof.

(a) The question is local (6.2), so assume $X = \operatorname{Spec} A$ for some Noetherian ring A, and $\mathcal{F} \cong \tilde{M}, \mathcal{G} \cong \tilde{N}$ for some A-modules M, N. Because A is Noetherian, there exists a finite free resolution of M

$$F_{\bullet} \longrightarrow M \longrightarrow 0$$

where each $F_{\bullet}$ is a free A-module of finite rank. Applying the $\tilde{}$-functor and setting $\mathcal{L}_{\bullet} = \tilde{F}_{\bullet}$, we obtain a free resolution of $\mathcal{F}$ of finite rank

$$\mathcal{L}_{\bullet} \longrightarrow \mathcal{F} \longrightarrow 0,$$

which we can use to compute $\mathbf{Ext}^i(\mathcal{F}, \mathcal{G})$ (6.5)

$$\mathbf{Ext}^i(\mathcal{F}, \mathcal{G}) \cong h^i(\mathbf{Hom}(\mathcal{L}_{\bullet}, \mathcal{G})).$$

By equivalence of categories between coherent $\mathcal{O}_X$-modules and finitely generated A-modules (§2, 5.5), one has

$$\mathbf{Hom}(\mathcal{L}_{\bullet}, \mathcal{G}) \cong \widetilde{\operatorname{Hom}_A(F_{\bullet}, N)}.$$

Each $\mathbf{Hom}(\mathcal{L}_{\bullet}, \mathcal{G})$ is coherent because each $\operatorname{Hom}(F_{\bullet}, N)$ is finitely generated, and A is Noetherian, so any sub-$\mathcal{O}_X$-module of $\mathbf{Hom}(\mathcal{L}_{\bullet}, \mathcal{G})$ is also coherent (§2, 5.7). Hence, $\mathbf{Ext}^i(\mathcal{F}, \mathcal{G})$ is coherent for every $i \geq 0$.

(b) The proof is essentially the same as (a) except $\mathcal{G} \cong \tilde{N}$ for some A-module N, and the result still follows by (§2, 5.7).

□

6. Let A be a regular local ring, and let M be a finitely generated A-module. In this case, strengthen the result (6.10A) as follows.

(a) M is projective if and only if $\operatorname{Ext}^i(M, A) = 0$ for all $i > 0$.

(b) Use (a) to show that for any n, $\operatorname{pd} M \leq n$ if and only if $\operatorname{Ext}^i(M, A) = 0$ for all $i > n$.

Proof.

(a) All noetherian local rings have finite Krull dimension, so $n = \operatorname{pd}(M) < \infty$ (6.11A), and $\operatorname{Ext}^i_A(M, N) = 0$ for all $i > n$ and all A-modules N (6.10A). Suppose $\operatorname{Ext}^i_A(M, A) = 0$ for all $i > 0$. Let N be any finitely generated A-module. There exists an exact sequence

$$0 \longrightarrow K \longrightarrow A^{\oplus m} \longrightarrow N \longrightarrow 0,$$

where K is some finitely generated A-module. Applying $\operatorname{Hom}_A(M, \cdot)$ gives a long exact sequence in cohomology

$$0 \longrightarrow \operatorname{Hom}_A(M, K) \longrightarrow \operatorname{Hom}_A(M, A^{\oplus m}) \longrightarrow \operatorname{Hom}_A(M, N) \longrightarrow$$

$$\longrightarrow \operatorname{Ext}^1_A(M, K) \longrightarrow \operatorname{Ext}^1_A(M, A^{\oplus m}) \longrightarrow \operatorname{Ext}^1_A(M, N) \longrightarrow \cdots$$

$$\cdots \longrightarrow \operatorname{Ext}^n_A(M, K) \longrightarrow \operatorname{Ext}^n_A(M, A^{\oplus m}) \longrightarrow \operatorname{Ext}^n_A(M, N) \longrightarrow 0.$$

The $\operatorname{Hom}_A(M, \cdot)$ functor commute with direct sums, so $\operatorname{Ext}^i_A(M, A^{\oplus m}) \cong \operatorname{Ext}^i_A(M, A)^{\oplus m} = 0$ for all $i > 0$. By the long exact sequence of above, $\operatorname{Ext}^{i-1}_A(M, N) \cong \operatorname{Ext}^i_A(M, K)$ and $\operatorname{Ext}^n_A(M, N) = 0$. Thus, $\operatorname{Ext}^n_A(M, N) = 0$ for all finitely generated A-modules N, and descending induction on i shows $\operatorname{Ext}^i_A(M, N) = 0$ for all $i > 0$ and all finitely generated A-modules N.
By assumption, M is finitely generated, so there exists an exact sequence

$$0 \longrightarrow Q \longrightarrow A^{\oplus r} \longrightarrow M \longrightarrow 0$$

where Q is a finitely generated A-module. Since $\operatorname{Ext}^1(M, Q) = 0$, which classifies all extensions of M by Q, $A^{\oplus r}$ must be the trivial extension of M by Q, i.e., $A^{\oplus r} \cong M \oplus Q$.
Let $\pi : N' \to N$ be a surjective A-module homomorphism and $\phi : M \to N$ any A-module homomorphism. There is a natural extension of ϕ to $A^{\oplus r}$ via taking the pull-back by the projection $A^{\oplus r} \to M$. Any free module is projective, so there exists unique A-module homomorphism $h : A^{\oplus r} \to N'$ such that $\phi' = \pi \circ h$, so by restriction one has $\phi = \pi \circ h\big|_M$. Hence, M is projective. In particular, a module is projective if and only if it is a direct summand of a free module.

(b) We prove the following general result from (Weibel, 1994):

Lemma 1. *Let $F : \mathcal{C} \to \mathcal{D}$ be a left exact contravariant functor between two Abelian categories. Assume $\mathcal{A}$ has enough projectives. If*

$$0 \longrightarrow N \longrightarrow P_m \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

is an exact sequence in $\mathcal{C}$ with the P_i projective (or F-cyclic), then $\mathrm{R}^i F(M) \cong \mathrm{R}^{i-m-1} F(N)$ for $i \geq m+2$.

For each $0 \le j \le m+1$, label the morphisms $f_j : P_j \to P_{j-1}$, where P_{-1}, P_{m+1} understood as M, N, respectively. We have an exact sequence

$$0 \longrightarrow \operatorname{im} f_{j+1} \longrightarrow P_j \longrightarrow \operatorname{im} f_j \longrightarrow 0,$$

which induces the natural isomorphism

$$\mathrm{R}^i F(\operatorname{im} f_j) \cong \mathrm{R}^{i-1} F(\operatorname{im} f_{j+1})$$

for all $i \ge 2$. Thus,

$$\mathrm{R}^i F(\operatorname{im} f_j) \cong \mathrm{R}^{i-k} F(\operatorname{im} f_{j+k})$$

for all $i \ge k+1$. Since $\operatorname{im} f_0 = M$ and $\operatorname{im} f_{m+1} = N$, setting $j = 0$ and $k = m+1$, we obtain our desired result.

Continuing on to our main problem, let M be a finitely generated A-module, and suppose $\operatorname{Ext}^i(M, A) = 0$ for all $i > n$. Let $P_\bullet \to M \to 0$ be any projective resolution of M, and consider the exact sequence

$$0 \longrightarrow N \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

where N is the image of $P_{n+1} \to P_n$. Applying Lemma 1 with $F = \operatorname{Hom}(\cdot, A)$ and $i = n+1$ so that $\mathrm{R}^i F = \operatorname{Ext}^i(\cdot, A)$, one has an isomorphism

$$\operatorname{Ext}^{n+1}(M, A) \cong \operatorname{Ext}^1(N, A).$$

But $\operatorname{Ext}^{n+1}(M, A) = 0$, so N is projective by (a). Hence, $\operatorname{pd} M \le n$.

□

7. Let $X = \operatorname{Spec} A$ be an affine Noetherian scheme. Let M, N be A-modules, with M finitely generated. Then

$$\operatorname{Ext}^i_X(\tilde{M}, \tilde{N}) \cong \operatorname{Ext}^i_A(M, N)$$

and

$$\mathbf{Ext}^i_X(\tilde{M}, \tilde{N}) \cong \widetilde{\operatorname{Ext}^i_A(M, N)}.$$

Proof. Both sides are δ-functors in N from $\mathfrak{Mod}(A)$ to $\mathfrak{Ab}$, and the functors $\operatorname{Hom}_X(\tilde{M}, \tilde{\cdot})$ and $\operatorname{Hom}_A(M, \cdot)$ are equal (§2, 5.5), so their derived functors (as functors from $\mathfrak{Mod}(A)$ to $\mathfrak{Ab}$) are the same.

For the next isomorphism, both sides are δ-functors in N from $\mathfrak{Mod}(A)$ to $\mathfrak{Mod}(X)$ because the $\tilde{\ }$-functor is an exact functor. Both sides vanish for $i > 0$ and N injective A-module because localization is exact, and $\tilde{N}$ is flasque if N is injective (3.4). Thus, to invoke (1.3A), we need to show the functors are equal for $i = 0$. We check locally. If $x \in X$ is a point, let $\mathfrak{p}$ be the prime ideal in A corresponding to x. By (§2, 5.1) and (6.8), we have

$$\mathbf{Ext}^i_X(\tilde{M}, \tilde{N})_x \cong \operatorname{Ext}^i_{A_\mathfrak{p}}(M_\mathfrak{p}, N_\mathfrak{p}), \quad \widetilde{\operatorname{Ext}^i_A(M, N)}_x \cong \operatorname{Ext}^i(M, N)_\mathfrak{p},$$

and there is a natural map of $\mathcal{O}_X$-modules

$$\begin{aligned} \widetilde{\operatorname{Hom}_A(M, N)} &\to \mathbf{Hom}_X(\tilde{M}, \tilde{N}) \\ f/s &\mapsto (m \mapsto f(m)/s). \end{aligned}$$

Thus, we reduce to the following algebraic problem: let A be a Noetherian ring, M, N A-modules with M finitely generated. Then the natural map of $A_\mathfrak{p}$-modules

$$\operatorname{Hom}_A(M, N)_\mathfrak{p} \to \operatorname{Hom}_{A_\mathfrak{p}}(M_\mathfrak{p}, N_\mathfrak{p})$$

is an isomorphism. If M is finitely generated, there exists an exact sequence

$$A^{\oplus n} \longrightarrow A^{\oplus r} \longrightarrow M \longrightarrow 0.$$

Applying $\operatorname{Hom}_A(\cdot, N)$ and localizing the first exact sequence gives

$$0 \longrightarrow \operatorname{Hom}_A(M, N)_\mathfrak{p} \longrightarrow \operatorname{Hom}_A(A^{\oplus r}, N)_\mathfrak{p} \longrightarrow \operatorname{Hom}_A(A^{\oplus n}, N)_\mathfrak{p}$$

Switching the order, i.e., localizing first then applying $\operatorname{Hom}_{A_\mathfrak{p}}(\cdot, N_\mathfrak{p})$, gives

$$0 \longrightarrow \operatorname{Hom}_{A_\mathfrak{p}}(M_\mathfrak{p}, N_\mathfrak{p}) \longrightarrow \operatorname{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus r}, N_\mathfrak{p}) \longrightarrow \operatorname{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus n}, N_\mathfrak{p}).$$

The statement obvious holds For finite free modules

$$\operatorname{Hom}_A(A^{\oplus r}, N)_\mathfrak{p} \cong N_\mathfrak{p}^{\oplus r} \cong \operatorname{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus r}, N_\mathfrak{p})$$

We obtain our desired isomorphism by applying the five lemma to the following commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \operatorname{Hom}_A(M, N)_\mathfrak{p} \arrow[r] \arrow[d] & \operatorname{Hom}_A(A^{\oplus r}, N)_\mathfrak{p} \arrow[r] \arrow[d] & \operatorname{Hom}_A(A^{\oplus n}, N)_\mathfrak{p} \arrow[d] \\ 0 \arrow[r] & \operatorname{Hom}_{A_\mathfrak{p}}(M_\mathfrak{p}, N_\mathfrak{p}) \arrow[r] & \operatorname{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus r}, N_\mathfrak{p}) \arrow[r] & \operatorname{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus n}, N_\mathfrak{p}) \end{tikzcd}.$$

□

8. Prove the following theorem of Kleiman: if X is a Noetherian, integral, separated, locally factorial scheme, then every coherent sheaf on X is a quotient of a locally free sheaf (of finite rank).

(a) First show that open sets of the form X_s, for various $s \in \Gamma(X, \mathcal{L})$, and various invertible sheaves $\mathcal{L}$ on X, form a base for the topology of X.

(b) Now use (II, 5.14) to show that any coherent sheaf is a quotient of a direct sum $\bigoplus \mathcal{L}_i^{n_i}$ for various invertible sheaves $\mathcal{L}_i$ and various integers n_i.

10. *Duality for a Finite Flat Morphism.*

(a) Let $f : X \to Y$ be a finite morphism of Noetherian schemes. For any quasi-coherent $\mathcal{O}_Y$-module $\mathcal{G}$, $\mathbf{Hom}_Y(f_*\mathcal{O}_X, \mathcal{G})$ is a quasi-coherent $f_*\mathcal{O}_X$-module, hence corresponds to a quasi-coherent $\mathcal{O}_X$-module, which we call $f^!\mathcal{G}$ (II, Ex. 5.17e).

(b) Show that for any coherent $\mathcal{F}$ on X and any quasi-coherent $\mathcal{G}$ on Y, there is a natural isomorphism

$$f_* \mathbf{Hom}_X(\mathcal{F}, f^!\mathcal{G}) \xrightarrow{\sim} \mathbf{Hom}_Y(f_*\mathcal{F}, \mathcal{G}).$$

(c) For each $i \geq 0$, there is a natural map

$$\varphi_i : \mathrm{Ext}^i_X(\mathcal{F}, f^!\mathcal{G}) \to \mathrm{Ext}^i_Y(f_*\mathcal{F}, \mathcal{G}).$$

(d) Now assume that X and Y are separated, $\mathfrak{Coh}(X)$ has enough locally frees, and assume that $f_*\mathcal{O}_X$ is locally free on Y (this is equivalent to saying f flat - see §9). Show that φ_i is an isomorphism for all i, all $\mathcal{F}$ coherent on X, and all $\mathcal{G}$ quasi-coherent on Y.

Proof.

(b) The question is local on Y, so we can assume $Y = \operatorname{Spec} A$ for some Noetherian ring A. Then $X = \operatorname{Spec} B$ for some finite A-algebra B so that

$$f_*\mathcal{O}_X \cong \tilde{B}, \quad \mathcal{F} \cong \tilde{M}, \quad \mathcal{G} \cong \tilde{N}$$

for some finitely generated B-module M, and A-module N. Now, the statement just says there is a natural isomorphism of A-modules

$$\mathrm{Hom}_B(M, \mathrm{Hom}_A(B, N)) \xrightarrow{\sim} \mathrm{Hom}_A(M, N).$$

(c) If $\mathcal{H}$ is any quasi-coherent $\mathcal{O}_X$-module, there is a natural map

$$\mathrm{Hom}_X(\mathcal{F}, \mathcal{H}) \to \mathrm{Hom}_Y(f_*\mathcal{F}, f_*\mathcal{H})$$

by functoriality of f_*. Note that f_* is an exact functor when f is an finite morphism (8.1), so both $\mathrm{Ext}^i_X(\mathcal{F}, \cdot)$ and $\mathrm{Ext}^i_Y(f_*\mathcal{F}, f_*\cdot)$ is a δ-functor in $\mathcal{H}$ from $\mathfrak{Mod}(X)$ to $\mathfrak{Ab}$. Since $\mathrm{Ext}^i_X(\mathcal{F}, \cdot)$ is a universal δ-functor (1.4), there is an induced natural map by definition

$$\psi_i : \mathrm{Ext}^i_X(\mathcal{F}, \mathcal{H}) \to \mathrm{Ext}^i_Y(f_*\mathcal{F}, f_*\mathcal{H}).$$

Now, consider the case $\mathcal{H} = f^!\mathcal{G}$. Taking the global sections of (b) shows

$$\mathrm{Hom}_X(f^!\mathcal{G}, f^!\mathcal{G}) \cong \mathrm{Hom}_Y(f_*f^!\mathcal{G}, \mathcal{G}),$$

so let $\xi : f_*f^!\mathcal{G} \to \mathcal{G}$ be the morphism corresponding to the identitiy $1_{f^!\mathcal{G}}$. Then ξ induces a map of derived functors

$$\xi_i : \mathrm{Ext}^i_Y(f_*\mathcal{F}, f_*f^!\mathcal{G}) \to \mathrm{Ext}_Y(f_*\mathcal{F}, \mathcal{G}).$$

Hence, $\varphi_i = \xi_i \circ \psi_i$ is our desired natural map.

(d)

$\square$